

## 1. Executive Summary

|                         |        | Traffic Summary        |  | Count      |  |
|-------------------------|--------|------------------------|--|------------|--|
| Metric                  | Value  | Total Flows Observed   |  | 132190     |  |
| Total Threats Mitigated | 25     | Total Packets Analyzed |  | 1652431434 |  |
| High Priority           | 5      | Malicious Packets      |  | 1651649174 |  |
| Low Priority            | 20     | True Negatives Passed  |  | 103769     |  |
| Manual Releases         | 0      |                        |  |            |  |
| Manual Blocks           | 0      | Attack Vector          |  | Count      |  |
|                         |        | ICMP Flood             |  | 0          |  |
| ML Metrics              | Value  | SYN Flood              |  | 15         |  |
| False Positives         | 127    | UDP Flood              |  | 2          |  |
| FP Rate                 | 0.08%  | Uncertain              |  | 8          |  |
| IF Precision            | 99.70% |                        |  |            |  |
| IF Recall               | 93.42% | Action                 |  | Count      |  |
| IF F1-Score             | 96.46% | Quarantined            |  | 5          |  |
| IF Accuracy             | 98.43% | Time Ban               |  | 4          |  |
|                         |        | Blackhole              |  | 0          |  |
|                         |        | Released               |  | 7          |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.70% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 93.42% | Share of actual attacks successfully flagged         |
| F1-Score                  | 96.46% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 98.43% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.08%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 6.58%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 93.42% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.92% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 28238          | FN 1989           |
| Actual: Normal | FP 84             | TN 101780         |

### 2b. Random Forest: Attack Classification

| Metric    | Value   | Description                                                  |
|-----------|---------|--------------------------------------------------------------|
| Precision | 100.00% | Share of classified flows assigned the correct attack type   |
| Recall    | 100.00% | Share of attacks of each type that were correctly identified |
| F1-Score  | 100.00% | Balanced measure combining Precision and Recall              |
| Accuracy  | 100.00% | Overall proportion of correct classifications                |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 28238          | 0               | 0              |
| Act: ICMP | 0              | 0               | 0              |
| Act: UDP  | 0              | 0               | 0              |

### 2c. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 413.1 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 5.5 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2d. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 1.14% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 4.54% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 6.90% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.17 | 3        | 3            | 2026-09-15 19:21:36 | 2026-09-15 19:22:58 |
| 10.0.0.16 | 2        | 2            | 2026-09-15 19:21:35 | 2026-09-15 19:22:46 |
| 10.0.0.18 | 2        | 2            | 2026-09-15 19:21:45 | 2026-09-15 19:22:56 |
| 10.0.0.19 | 2        | 2            | 2026-09-15 19:21:38 | 2026-09-15 19:22:49 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector    | Confidence | Priority | Action        |
|---------------------|-----------|-------|-----------|------------|----------|---------------|
| 2026-09-15 10:06:39 | 10.0.0.17 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:06:40 | 10.0.0.16 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:06:48 | 10.0.0.18 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:06:48 | 10.0.0.19 | DDoS  | Uncertain | -          | Low      | Quarantined   |
| 2026-09-15 10:06:49 | 10.0.0.17 | DDoS  | SYN Flood | 91.8%      | Critical | Time Ban (1m) |
| 2026-09-15 10:06:50 | 10.0.0.17 | DDoS  | SYN Flood | 91.8%      | Critical | Time Ban      |
| 2026-09-15 10:06:50 | 10.0.0.16 | DDoS  | SYN Flood | 93.3%      | Critical | Time Ban (1m) |
| 2026-09-15 10:06:50 | 10.0.0.16 | DDoS  | SYN Flood | 93.3%      | Critical | Time Ban      |
| 2026-09-15 10:06:58 | 10.0.0.18 | DDoS  | SYN Flood | 90.9%      | Critical | Time Ban (1m) |
| 2026-09-15 10:06:58 | 10.0.0.19 | DDoS  | SYN Flood | 90.9%      | Critical | Time Ban (1m) |
| 2026-09-15 10:06:58 | 10.0.0.18 | DDoS  | SYN Flood | 90.9%      | Critical | Time Ban      |
| 2026-09-15 10:07:00 | 10.0.0.19 | DDoS  | SYN Flood | 91.8%      | Critical | Time Ban      |
| 2026-09-15 10:07:50 | 10.0.0.17 | DDoS  | SYN Flood | 91.8%      | High     | Released      |
| 2026-09-15 10:07:51 | 10.0.0.16 | DDoS  | SYN Flood | 93.3%      | Critical | Time Ban (2m) |
| 2026-09-15 10:07:59 | 10.0.0.18 | DDoS  | SYN Flood | 90.9%      | High     | Released      |
| 2026-09-15 10:07:59 | 10.0.0.19 | DDoS  | SYN Flood | 91.8%      | Critical | Time Ban (2m) |
| 2026-09-15 10:08:00 | 10.0.0.17 | DDoS  | SYN Flood | 93.2%      | Critical | Time Ban (2m) |
| 2026-09-15 10:08:09 | 10.0.0.18 | DDoS  | SYN Flood | 91.4%      | Critical | Time Ban (2m) |
| 2026-09-15 10:13:28 | 10.0.0.6  | DDoS  | UDP Flood | 92.4%      | High     | Time Ban (1m) |
| 2026-09-15 10:13:34 | 10.0.0.5  | DDoS  | Uncertain | 59.1%      | Low      | Released      |
| 2026-09-15 10:13:39 | 10.0.0.1  | DDoS  | Uncertain | 58.9%      | Low      | Released      |
| 2026-09-15 10:14:29 | 10.0.0.6  | DDoS  | UDP Flood | 92.4%      | High     | Released      |
| 2026-09-15 10:14:58 | 10.0.0.6  | DDoS  | Uncertain | 48.4%      | Low      | Quarantined   |
| 2026-09-15 10:18:52 | 10.0.0.19 | DDoS  | SYN Flood | 93.4%      | High     | Released      |
| 2026-09-15 10:23:40 | 10.0.0.2  | DDoS  | Uncertain | 58.9%      | Low      | Released      |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.17 | SYN Flood | 0.6165   | 92.6% | Critical | Phase 2 | 3        | 11s      | 2026-09-15 19:22:58 | Attack persisted - escalating to time ban |
| 10.0.0.18 | SYN Flood | 0.6026   | 93.1% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-15 19:22:56 | Attack persisted - escalating to time ban |
| 10.0.0.19 | SYN Flood | 0.6365   | 92.8% | Critical | Phase 2 | 2        | 1m 11s   | 2026-09-15 19:22:49 | Attack persisted - escalating to time ban |
| 10.0.0.17 | SYN Flood | 0.5871   | 92.6% | High     | Phase 2 | 1        | 1m 11s   | 2026-09-15 19:22:47 | Ban Expired                               |

| Source IP | Vector    | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|-----------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.16 | SYN Flood | 0.5985   | 93.4% | High     | Phase 2 | 2        | 1m 11s   | 2026-09-15 19:22:46 | Attack persisted - escalating to time ban |
| 10.0.0.18 | SYN Flood | 0.6421   | 93.1% | Critical | Phase 2 | 1        | 10s      | 2026-09-15 19:21:55 | Attack persisted - escalating to time ban |
| 10.0.0.19 | SYN Flood | 0.6456   | 90.1% | Critical | Phase 2 | 1        | 11s      | 2026-09-15 19:21:49 | Attack persisted - escalating to time ban |
| 10.0.0.17 | SYN Flood | 0.6342   | 92.6% | Critical | Phase 2 | 1        | 11s      | 2026-09-15 19:21:47 | Attack persisted - escalating to time ban |
| 10.0.0.16 | SYN Flood | 0.6362   | 93.4% | Critical | Phase 2 | 1        | 11s      | 2026-09-15 19:21:46 | Attack persisted - escalating to time ban |

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE